

EC400: Dynamic Optimization

DMITRY MUKHIN
d.mukhin@lse.ac.uk

Introduction

- This part of the course is about solving **dynamic problems**

Introduction

- This part of the course is about solving **dynamic problems**
- Some examples of dynamic models:
 - ① *savings problem*: consumption today vs. tomorrow
 - ② *investment problem*: dividends today vs. tomorrow
 - ③ *portfolio choice*: returns vs. volatility
 - ④ *sticky prices*: profits vs. menu costs
 - ⑤ *job search*: wage vs. time
 - ⑥ *public finance*: taxes vs. debt
 - ⑦ ...

Introduction

- This part of the course is about solving **dynamic problems**
- Some examples of dynamic models:
 - ① *savings problem*: consumption today vs. tomorrow
 - ② *investment problem*: dividends today vs. tomorrow
 - ③ *portfolio choice*: returns vs. volatility
 - ④ *sticky prices*: profits vs. menu costs
 - ⑤ *job search*: wage vs. time
 - ⑥ *public finance*: taxes vs. debt
 - ⑦ ...
- Limitations of static methods $\Rightarrow$ alternative approaches

Introduction

- Classification of problems and methods:
 - ① **deterministic** vs. **stochastic** setup
 - ② **discrete** vs. **continuous** time
 - ③ **sequential** vs. **recursive** approach

Introduction

- Classification of problems and methods:
 - ① **deterministic** vs. **stochastic** setup
 - ② **discrete** vs. **continuous** time
 - ③ **sequential** vs. **recursive** approach
- “Cookbook” approach:
 - tools to solve models
 - heuristic proofs and derivations
 - macroeconomic examples

BASELINE MODEL

Finite-Horizon Problem

- Household consumption-saving problem:

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}_{t=0}^{t=T}} \quad & \sum_{t=0}^T \beta^t u(C_t) \\ \text{s.t.} \quad & B_{t+1} = RB_t + Y_t - C_t \\ & B_{T+1} \geq 0, \quad B_0 \text{ is given} \end{aligned}$$

Finite-Horizon Problem

- Household consumption-saving problem:

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}_{t=0}^{t=T}} \quad & \sum_{t=0}^T \beta^t u(C_t) \\ \text{s.t.} \quad & B_{t+1} = RB_t + Y_t - C_t \\ & B_{T+1} \geq 0, \quad B_0 \text{ is given} \end{aligned}$$

- Simplifying assumptions:
 - finite horizon
 - no uncertainty
 - one good, exogenous income $\{Y_t\}$, constant $R_t = R$, etc.

Finite-Horizon Problem

- Household consumption-saving problem:

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}_{t=0}^T} \quad & \sum_{t=0}^T \beta^t u(C_t) \\ \text{s.t.} \quad & B_{t+1} = RB_t + Y_t - C_t \\ & B_{T+1} \geq 0, \quad B_0 \text{ is given} \end{aligned}$$

- Simplifying assumptions:

- finite horizon
- no uncertainty
- one good, exogenous income $\{Y_t\}$, constant $R_t = R$, etc.

- Preferences:

- discount factor $\beta \in [0, 1)$
- utility function $u(\cdot)$ is increasing and concave
- $\lim_{C \rightarrow 0} u'(C) = \infty$

Direct Approach

- How do we approach such problem?

Direct Approach

- How do we approach such problem?
- Can reinterpret it as a high-dimensional static problem:

$$\max_{C_0 \dots C_T, B_1 \dots B_{T+1}} U(C_0, C_1, \dots, C_T)$$

$$\text{s.t. } C_0 = RB_0 + Y_0 - B_1$$

...

$$C_T = RB_T + Y_T - B_{T+1}$$

$$B_{T+1} \geq 0$$

Direct Approach

- How do we approach such problem?
- Can reinterpret it as a high-dimensional static problem:

$$\max_{C_0 \dots C_T, B_1 \dots B_{T+1}} U(C_0, C_1, \dots, C_T)$$

$$\text{s.t. } C_0 = RB_0 + Y_0 - B_1$$

...

$$C_T = RB_T + Y_T - B_{T+1}$$

$$B_{T+1} \geq 0$$

- Can apply the optimization tools for static problems:
 - ① write down the Lagrangian
 - ② derive the Kuhn-Tucker conditions
 - ③ substitute out the Lagrange multipliers
 - ④ solve $2 \times (T + 1)$ equations with $2 \times (T + 1)$ unknowns

Lagrangian

$$\mathcal{L} = \sum_{t=0}^T \beta^t u(C_t) + \sum_{t=0}^T \lambda_t \left(RB_t + Y_t - C_t - B_{t+1} \right) + \mu B_{T+1},$$

Optimality Conditions

① Euler equation (EE):

$$u'(C_t) = \beta R u'(C_{t+1}), \quad \forall t < T$$

Optimality Conditions

① Euler equation (EE):

$$u'(C_t) = \beta R u'(C_{t+1}), \quad \forall t < T$$

② Transversality condition (TVC):

$$\beta^T u'(C_T) B_{T+1} = 0$$

Optimality Conditions

① Euler equation (EE):

$$u'(C_t) = \beta R u'(C_{t+1}), \quad \forall t < T$$

② Transversality condition (TVC):

$$\beta^T u'(C_T) B_{T+1} = 0$$

③ Resource (budget) constraint (BC):

$$B_{t+1} = RB_t + Y_t - C_t$$

$\Rightarrow$ optimal path $\{C_t, B_{t+1}\}_{t=0}^{t=T}$ with $2 \times (T + 1)$ unknowns is determined by $T + 1$ BC's, T EE's, and one TVC

Example

- Assume log preferences: $u(C) = \log C$

Example

- Assume log preferences: $u(C) = \log C$
- Equilibrium system:

$$C_{t+1} = \beta R C_t \quad (\text{EE})$$

$$B_{t+1} = R B_t + Y_t - C_t \quad (\text{RC})$$

$$B_{T+1} = 0 \quad (\text{TVC})$$

Example

- Assume log preferences: $u(C) = \log C$
- Equilibrium system:

$$C_{t+1} = \beta RC_t \quad (\text{EE})$$

$$B_{t+1} = RB_t + Y_t - C_t \quad (\text{RC})$$

$$B_{T+1} = 0 \quad (\text{TVC})$$

- (RC) + (TVC) $\Rightarrow$ **intertemporal** budget constraint:

$$\sum_{t=0}^T R^{-t} C_t = RB_0 + \sum_{t=0}^T R^{-t} Y_t$$

Example

- Assume log preferences: $u(C) = \log C$
- Equilibrium system:

$$C_{t+1} = \beta R C_t \quad (\text{EE})$$

$$B_{t+1} = R B_t + Y_t - C_t \quad (\text{RC})$$

$$B_{T+1} = 0 \quad (\text{TVC})$$

- (RC) + (TVC) $\Rightarrow$ **intertemporal** budget constraint:

$$\sum_{t=0}^T R^{-t} C_t = R B_0 + \sum_{t=0}^T R^{-t} Y_t$$

- Substitute in (EE) and solve:

$$C_0 = \frac{R B_0 + \sum_{t=0}^T R^{-t} Y_t}{\sum_{t=0}^T \beta^t}$$

INFINITE HORIZON

Infinite-Horizon Problem

- Household problem with infinite horizon:

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t u(C_t) \\ \text{s.t.} \quad & B_{t+1} = RB_t + Y_t - C_t \\ & B_t \geq -\bar{B}, \quad B_0 \text{ is given} \end{aligned}$$

- **no-Ponzi game condition (NPGC)** excludes trivial solutions

Infinite-Horizon Problem

- Household problem with infinite horizon:

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t u(C_t) \\ \text{s.t.} \quad & B_{t+1} = RB_t + Y_t - C_t \\ & B_t \geq -\bar{B}, \quad B_0 \text{ is given} \end{aligned}$$

— **no-Ponzi game condition (NPGC)** excludes trivial solutions

- Lagrangian:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(C_t) + \sum_{t=0}^{\infty} \lambda_t (RB_t + Y_t - C_t - B_{t+1})$$

Infinite-Horizon Problem

- Household problem with infinite horizon:

$$\begin{aligned} \max_{\{C_t, B_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t u(C_t) \\ \text{s.t.} \quad & B_{t+1} = RB_t + Y_t - C_t \\ & B_t \geq -\bar{B}, \quad B_0 \text{ is given} \end{aligned}$$

— **no-Ponzi game condition (NPGC)** excludes trivial solutions

- Lagrangian:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(C_t) + \sum_{t=0}^{\infty} \lambda_t (RB_t + Y_t - C_t - B_{t+1})$$

- The optimality conditions:

$$u'(C_t) = \beta R u'(C_{t+1}) \quad (\text{EE})$$

$$\lim_{T \rightarrow \infty} \beta^T u'(C_T) B_{T+1} = 0 \quad (\text{TVC})$$

Example

- Consider again log preferences: $u(C) = \log C$

Example

- Consider again log preferences: $u(C) = \log C$
- Equilibrium system:

$$C_{t+1} = \beta R C_t \quad (\text{EE})$$

$$B_{t+1} = R B_t + Y_t - C_t \quad (\text{RC})$$

$$\lim_{T \rightarrow \infty} \frac{\beta^T}{C_T} B_{T+1} = 0 \quad (\text{TVC})$$

Example

- Consider again log preferences: $u(C) = \log C$
- Equilibrium system:

$$C_{t+1} = \beta R C_t \quad (\text{EE})$$

$$B_{t+1} = R B_t + Y_t - C_t \quad (\text{RC})$$

$$\lim_{T \rightarrow \infty} \frac{\beta^T}{C_T} B_{T+1} = 0 \quad (\text{TVC})$$

- (EE) + (TVC) + (RC) $\Rightarrow$ intertemporal budget constraint

$$\sum_{t=0}^{\infty} R^{-t} C_t = R B_0 + \sum_{t=0}^{\infty} R^{-t} Y_t$$

Example

- Consider again log preferences: $u(C) = \log C$
- Equilibrium system:

$$C_{t+1} = \beta R C_t \quad (\text{EE})$$

$$B_{t+1} = R B_t + Y_t - C_t \quad (\text{RC})$$

$$\lim_{T \rightarrow \infty} \frac{\beta^T}{C_T} B_{T+1} = 0 \quad (\text{TVC})$$

- (EE) + (TVC) + (RC) $\Rightarrow$ intertemporal budget constraint

$$\sum_{t=0}^{\infty} R^{-t} C_t = R B_0 + \sum_{t=0}^{\infty} R^{-t} Y_t$$

- Substitute in (EE) and solve:

$$C_0 = (1 - \beta) \left(R B_0 + \sum_{t=0}^{\infty} R^{-t} Y_t \right)$$

CONTINUOUS TIME

Continuous Time Approach

- Some dynamic problems are more tractable in continuous time
- Heuristic derivations to show analogy at the level of
 - optimization problem
 - Lagrangian / Hamiltonian
 - optimality conditions
 - closed-form solution

Continuous Time Approach

- Some dynamic problems are more tractable in continuous time
- Heuristic derivations to show analogy at the level of
 - optimization problem
 - Lagrangian / Hamiltonian
 - optimality conditions
 - closed-form solution
- Rewrite conditions for time period of Δ and take limit $\Delta \rightarrow 0$
- Crucial to distinguish two types of variables:
 - ① **stocks** measured **at** a given time: capital, debt, wealth...
 - ② **flows** measured **over** a period of time: interest, GDP...

Savings Problem in Continuous Time

- Household budget constraint:
 - rewrite for period of length Δ :

$$B_{t+\Delta} = (1 + r\Delta)B_t + Y_t\Delta - C_t\Delta$$

Savings Problem in Continuous Time

- Household budget constraint:
 - rewrite for period of length Δ :

$$B_{t+\Delta} = (1 + r\Delta)B_t + Y_t\Delta - C_t\Delta$$

- convert stocks into flows:

$$B_{t+\Delta} - B_t = rB_t\Delta + Y_t\Delta - C_t\Delta$$

Savings Problem in Continuous Time

- Household budget constraint:

- rewrite for period of length Δ :

$$B_{t+\Delta} = (1 + r\Delta)B_t + Y_t\Delta - C_t\Delta$$

- convert stocks into flows:

$$B_{t+\Delta} - B_t = rB_t\Delta + Y_t\Delta - C_t\Delta$$

- take limit $\Delta \rightarrow 0$:

$$\dot{B}_t = rB_t + Y_t - C_t, \quad \dot{B}_t \equiv \lim_{\Delta \rightarrow 0} \frac{B_{t+\Delta} - B_t}{\Delta}$$

Savings Problem in Continuous Time

- Household budget constraint:

- rewrite for period of length Δ :

$$B_{t+\Delta} = (1 + r\Delta)B_t + Y_t\Delta - C_t\Delta$$

- convert stocks into flows:

$$B_{t+\Delta} - B_t = rB_t\Delta + Y_t\Delta - C_t\Delta$$

- take limit $\Delta \rightarrow 0$:

$$\dot{B}_t = rB_t + Y_t - C_t, \quad \dot{B}_t \equiv \lim_{\Delta \rightarrow 0} \frac{B_{t+\Delta} - B_t}{\Delta}$$

- Utility:

- define $\rho > 0$ s.t. $\beta \equiv e^{-\rho\Delta}$

Savings Problem in Continuous Time

- Household budget constraint:

- rewrite for period of length Δ :

$$B_{t+\Delta} = (1 + r\Delta)B_t + Y_t\Delta - C_t\Delta$$

- convert stocks into flows:

$$B_{t+\Delta} - B_t = rB_t\Delta + Y_t\Delta - C_t\Delta$$

- take limit $\Delta \rightarrow 0$:

$$\dot{B}_t = rB_t + Y_t - C_t, \quad \dot{B}_t \equiv \lim_{\Delta \rightarrow 0} \frac{B_{t+\Delta} - B_t}{\Delta}$$

- Utility:

- define $\rho > 0$ s.t. $\beta \equiv e^{-\rho\Delta}$

- replace sum with integral

$$\sum_{t=0}^{\infty} e^{-\rho t} u(C_t) \Delta \longrightarrow \int_0^{\infty} e^{-\rho t} u(C_t) dt$$

HAMILTONIAN

Optimal Control Problem

- Consider generalized continuous-time problem:

$$\begin{aligned} \max_{\{C_t, B_t\}} & \int_0^{\infty} e^{-\rho t} u(C_t) dt \\ \text{s.t. } & \dot{B}_t = g(B_t, C_t) \end{aligned}$$

Optimal Control Problem

- Consider generalized continuous-time problem:

$$\begin{aligned} \max_{\{C_t, B_t\}} & \int_0^{\infty} e^{-\rho t} u(C_t) dt \\ \text{s.t. } & \dot{B}_t = g(B_t, C_t) \end{aligned}$$

- Define the **Hamiltonian**:

$$\mathcal{H}_t \equiv e^{-\rho t} u(C_t) + \lambda_t g(B_t, C_t)$$

Optimal Control Problem

- Consider generalized continuous-time problem:

$$\begin{aligned} \max_{\{C_t, B_t\}} \int_0^\infty e^{-\rho t} u(C_t) dt \\ \text{s.t. } \dot{B}_t = g(B_t, C_t) \end{aligned}$$

- Define the **Hamiltonian**:

$$\mathcal{H}_t \equiv e^{-\rho t} u(C_t) + \lambda_t g(B_t, C_t)$$

- Maximum principle:** necessary conditions for optimality are

$$\frac{\partial \mathcal{H}_t}{\partial C_t} = 0, \quad \frac{\partial \mathcal{H}_t}{\partial B_t} = -\dot{\lambda}_t, \quad \lim_{t \rightarrow \infty} \lambda_t B_t = 0$$

Optimal Control Problem

- Consider generalized continuous-time problem:

$$\begin{aligned} \max_{\{C_t, B_t\}} \int_0^\infty e^{-\rho t} u(C_t) dt \\ \text{s.t. } \dot{B}_t = g(B_t, C_t) \end{aligned}$$

- Define the **Hamiltonian**:

$$\mathcal{H}_t \equiv e^{-\rho t} u(C_t) + \lambda_t g(B_t, C_t)$$

- Maximum principle**: necessary conditions for optimality are

$$\frac{\partial \mathcal{H}_t}{\partial C_t} = 0, \quad \frac{\partial \mathcal{H}_t}{\partial B_t} = -\dot{\lambda}_t, \quad \lim_{t \rightarrow \infty} \lambda_t B_t = 0$$

- Intuition**: Lagrangian for a discrete-time version of the problem:

$$\mathcal{L} = \sum_{t=0}^{\infty} \left[\mathcal{H}_t \Delta - \lambda_t (B_{t+\Delta} - B_t) \right]$$

Optimal Control Problem

- Consider generalized continuous-time problem:

$$\begin{aligned} \max_{\{C_t, B_t\}} \int_0^\infty e^{-\rho t} u(C_t) dt \\ \text{s.t. } \dot{B}_t = g(B_t, C_t) \end{aligned}$$

- Define the **Hamiltonian**:

$$\mathcal{H}_t \equiv e^{-\rho t} u(C_t) + \lambda_t g(B_t, C_t)$$

- Maximum principle**: necessary conditions for optimality are

$$\frac{\partial \mathcal{H}_t}{\partial C_t} = 0, \quad \frac{\partial \mathcal{H}_t}{\partial B_t} = -\dot{\lambda}_t, \quad \lim_{t \rightarrow \infty} \lambda_t B_t = 0$$

- Optimality conditions:

$$\begin{aligned} \frac{\partial \mathcal{H}_t}{\partial C_t} \Delta = 0 & \Rightarrow \frac{\partial \mathcal{H}_t}{\partial C_t} = 0 \\ \frac{\partial \mathcal{H}_{t+\Delta}}{\partial B_{t+\Delta}} \Delta - \lambda_t + \lambda_{t+\Delta} = 0 & \Rightarrow \frac{\partial \mathcal{H}_t}{\partial B_t} = -\dot{\lambda}_t \end{aligned}$$

Example

- Assume log utility $u(C) = \log C$ and constant income $Y_t = Y$

Example

- Assume log utility $u(C) = \log C$ and constant income $Y_t = Y$
- Household problem:

$$\begin{aligned} \max_{\{C_t, B_t\}_{t=0}^{\infty}} & \int_0^{\infty} e^{-\rho t} \log C_t dt \\ \text{s.t. } & \dot{B}_t = rB_t + Y_t - C_t \end{aligned}$$

Example

- Assume log utility $u(C) = \log C$ and constant income $Y_t = Y$
- Household problem:

$$\begin{aligned} \max_{\{C_t, B_t\}_{t=0}^{\infty}} & \int_0^{\infty} e^{-\rho t} \log C_t dt \\ \text{s.t. } & \dot{B}_t = rB_t + Y_t - C_t \end{aligned}$$

- Hamiltonian:

$$\mathcal{H}_t \equiv e^{-\rho t} \log C_t + \lambda_t (rB_t + Y_t - C_t)$$

Example

- Assume log utility $u(C) = \log C$ and constant income $Y_t = Y$
- Household problem:

$$\begin{aligned} \max_{\{C_t, B_t\}_{t=0}^{\infty}} & \int_0^{\infty} e^{-\rho t} \log C_t dt \\ \text{s.t. } & \dot{B}_t = rB_t + Y_t - C_t \end{aligned}$$

- Hamiltonian:

$$\mathcal{H}_t \equiv e^{-\rho t} \log C_t + \lambda_t (rB_t + Y_t - C_t)$$

- Optimality conditions:

$$\begin{aligned} 0 &= \frac{\partial \mathcal{H}_t}{\partial C_t} = e^{-\rho t} \frac{1}{C_t} - \lambda_t \\ -\dot{\lambda}_t &= \frac{\partial \mathcal{H}_t}{\partial B_t} = r\lambda_t \\ \lim_{t \rightarrow \infty} \lambda_t B_t &= 0 \end{aligned}$$

Example

- Combine optimality conditions to get the Euler equation:

$$\frac{\dot{C}_t}{C_t} = r - \rho$$

Example

- Combine optimality conditions to get the Euler equation:

$$\frac{\dot{C}_t}{C_t} = r - \rho$$

- Solve this differential equation:

$$C_t = C_0 e^{(r-\rho)t}$$

Example

- Combine optimality conditions to get the Euler equation:

$$\frac{\dot{C}_t}{C_t} = r - \rho$$

- Solve this differential equation:

$$C_t = C_0 e^{(r-\rho)t}$$

- Integrate the flow budget constraint:

$$e^{-rt} B_t = B_0 + \int_0^t (Y - C_s) e^{-rs} ds$$

Example

- Combine optimality conditions to get the Euler equation:

$$\frac{\dot{C}_t}{C_t} = r - \rho$$

- Solve this differential equation:

$$C_t = C_0 e^{(r-\rho)t}$$

- Integrate the flow budget constraint:

$$e^{-rt} B_t = B_0 + \int_0^t (Y - C_s) e^{-rs} ds$$

- Use the transversality condition:

$$B_0 + \int_0^{\infty} (Y - C_s) e^{-rs} ds = 0$$

Example

- Combine optimality conditions to get the Euler equation:

$$\frac{\dot{C}_t}{C_t} = r - \rho$$

- Solve this differential equation:

$$C_t = C_0 e^{(r-\rho)t}$$

- Integrate the flow budget constraint:

$$e^{-rt} B_t = B_0 + \int_0^t (Y - C_s) e^{-rs} ds$$

- Use the transversality condition:

$$B_0 + \int_0^\infty (Y - C_s) e^{-rs} ds = 0$$

- Substitute in C_t and integrate:

$$C_0 = \rho \left(B_0 + \frac{1}{r} Y \right).$$